



Habib University School of Science & Engineering

Course:	Linear Integrated Circuits
Semester	Fall 2025
Quiz #	1
Date, Time :	October 16, 2025, 1 hour
Instructor :	Syed Arsalan Jawed
Total Marks :	100

Instructions for Students

1. Provide your answer within the empty spaces given after each question.
2. Keep your answers concise and to the point. Provide neat and tidy calculations and plots.
3. Support your answer with equations and simulation where asked and needed.

Question # 1 : [CLO-1,2,3]

[50]

The following circuits show a closed amplifier configuration using an opamp. The offset of the input differential is referred to the input as V_{os} .

- 1) Derive the expression of the closed loop Gains for both circuits assuming that the open loop gain $= 10^6$. $\{A_o = 1 \times 10^6\}$
- 2) If $V_{in}=0$ and $R_1=0$ and opamp is ideal and $V_{os}=1mV$, $V_{out}=?$ Consider $V_{CC}=15V$ and $V_{EE}=-15V$.
- 3) Simulate the circuit in XSCHEME/NGSPICE and attach schematic snap and waveform snap with hand-written annotations to explain the functioning.

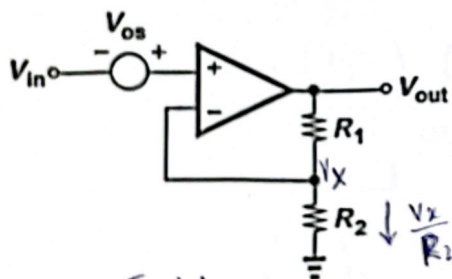


Fig 1.1.

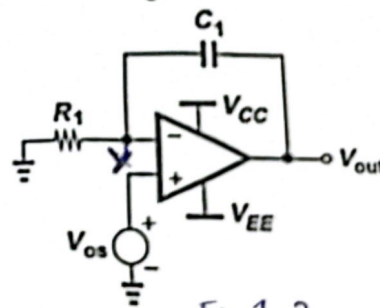


Fig 1.2

Q1.

1) $V_{in}=0$, $R_1=0$, $V_{os}=1mV$, $V_{CC}=15V$, $V_{EE}=-15V$, opamp ideal

Fig 1.1:

$$V_{out} = A_o (V_{in} + V_{os} - V_x)$$

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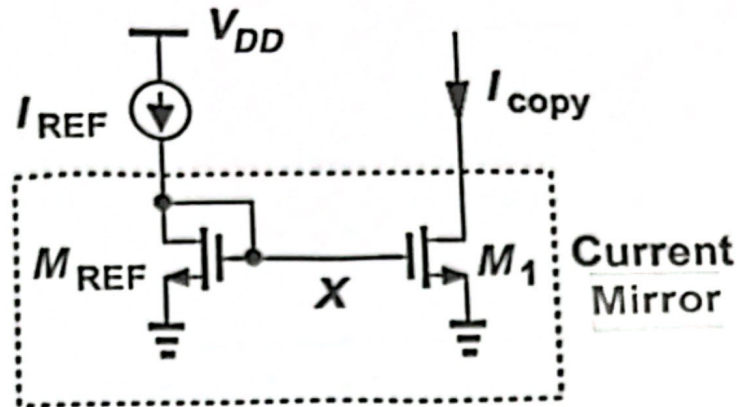
$$V_{out} = A_o (V_{os} - V_x + V_{in})$$

Fig 1.2

$$V_{out} = A_o (V_{os} - V_x)$$

For the following circuit, consider $W/L_{MREF} = 10\mu/1\mu$ and $W/L_1 = 100\mu/1\mu$, $I_{REF} = 10\mu A$, $V_{DD} = 1.8V$, $V_{TH} = 0.5V$. Consider the devices are following square law in strong inversion.

- 1) Derive the expression for I_{copy} considering there is no channel length modulation.
- 2) Derive the expression for V_X .
- 3) Take Home : Simulate Circuits in XSCHEME/NGSPICE and submit schematic and waveform snaps by October 16, 2300.



$$1) \quad \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_X - V_{TH})^2 = -I_{REF}$$

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{REF} (V_X - V_{TH})^2$$

$$I_{copy} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_X - V_{TH})^2$$

considering V_{TH} are equal.

$$I_{copy} = \frac{\left(\frac{W}{L} \right)_1}{\left(\frac{W}{L} \right)_{REF}} \cdot I_{REF}$$

$$I_{copy} = \frac{100\mu/1\mu}{10\mu/1\mu} \cdot 10\mu A$$

$$\boxed{I_{copy} = 100\mu A}$$

$$2) \quad V_X = \sqrt{\frac{2I_{REF}}{\mu_n C_{ox} \left(\frac{W}{L} \right)_1}} + V_{TH1}$$